Determining Cluster Membership Probability in Open Cluster M23

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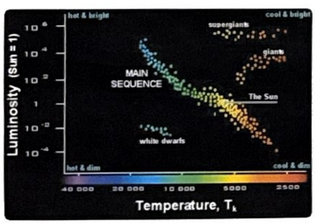
Introduction

Dr. Wilkerson and his students have been acquiring images of the field of view of open star cluster M23 for about 20 years at Luther College. The stars in open cluster M23 are all near one another, and are therefore all relatively the same distance from us. However, there are other stars that show up in this field of view that aren't in the cluster. We want to determine the likelihood that each star in the field of view is actually a member of open cluster M23.

Background

While most of the data is unfiltered, some images are taken through color filters, from which we can build color-magnitude diagrams. Color-magnitude diagrams are like Hertzberg-Russell (HR) diagrams but with apparent magnitudes instead of absolute magnitudes. There is a known pattern of main sequence stars on HR diagrams. A color-magnitude diagram shows the same pattern when it contains cluster stars that are all the same distance away. Other stars in the field create noise in the plot and appear randomly distributed. We can use these diagrams to estimate cluster membership probability for stars in the field.



Example HR Diagram

Image credit: https://www.wwu.edu/astro101/a101_hrdiagram.shtml

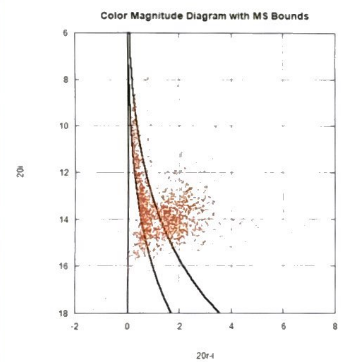
80-90% of stars plotted on HR Diagrams fall on the main sequence. As a star ages, it goes through a "main sequence life". Once a star has fused all the hydrogen in its core, it will leave the main sequence. Higher mass stars live much shorter main sequence lives and tend to evolve into red giants. Lower mass stars live longer main sequence lives and tend to evolve into white dwarfs.

Many other cluster membership publications used a statistical approach. One approach is to assign a proximity rating to stars based on their XY position in the field of view by assuming the cluster is centered in that field. This is one approach we will explore. Other approaches included the use of proper motion data, Bayesian inference, and random forest machine learning.

Plotting Color-Magnitude Diagram

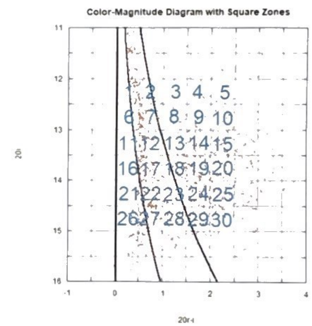
Using the filtered data acquired on 08-20-2006, we create a Color-Magnitude Diagram with an "R-I" color index and an "I" apparent magnitude. We observe a potential main sequence trend and superimpose a power-series fit on our plot. In order to obtain a "width" of the main sequence, we translate the original power-series curve to the left and right until the stars we are most confident in being cluster members are within the left and right bounds of the fit. We called this zone "MS Bounds" which is visible on our Color-Magnitude plot.

Two primary areas of interest include the very top of our MS Bounds curve, where some of those stars seem to be curving back to a redder color. This can be explained by the older stars essentially "leaving" the main sequence as they age, turning into red giant stars.



Estimating Cluster Probability

To begin assessing probability of cluster membership, we split our Color-Magnitude plot into 30 square zones, each with dimensions of 0.5 "R-I" and 0.5 "I". As the main sequence approaches the noisy portion of the plot, we utilize the MS bounds by calculating what fraction of the area of each square zone is occupied by the fit. This fraction gives us a rough idea about cluster membership probability (specifically for square zones that have a uniform distribution of stars), and provides us a "base" probability. For example, if the MS bound occupies half of a square zone, that zone is assigned a 50% base probability. In order to make these estimates realistic, a reasonable method to account for "noise stars" that are not cluster members, but appear in the field-of-view of M23. The noise profile is most dense where I=13-15, making stars in this range less likely to be on the main sequence. We identify a rough distribution of noise in relation to "I" value and apply an appropriate reduction of the base probability by for each range. The noisiest region (13.5-14) receives the maximum deduction of 50%.

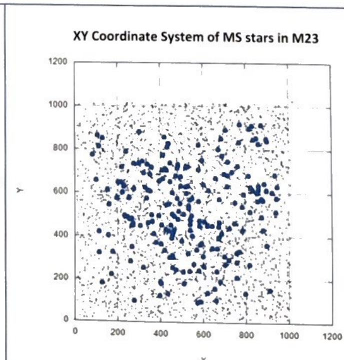


Non-uniformly populated zones must be assessed differently, for example zones 1, 6, 11, and 16. Each of these zones are positioned on the left edge of our main sequence fit. There is significantly less noise on the left side of the grid. We have confidence in our fit of the main sequence up to about I=14, and the stars that do fall in these zones seem to be right along this fit. Therefore, even though the fit only slightly passes through these zones, the stars in them are given high cluster membership probability.

To add our additional measure of positioning within the field, each star was assigned a score based on the density of main sequence stars in its respective grid below. The most populated grid contains 28 main sequence stars, and each after that was assigned a probability that was a fraction of 28. While these aren't very useful as stand-alone cluster probabilities, they will serve to help confirm the ones found using the main sequence bounds.

Position of Cluster Stars

As an extra measure to test how confident our calculations are, we check the XY position of stars in the field. The plot to the right lays coordinates of XY positions on our field images, with the stars we're most confident are main sequence from above bolded in blue. While there is not a distinct zone where a majority of stars are located, there are definitely zones with less stars (like most of the corners). There are also zones where the main sequence stars are more densely populated (right in the middle). Therefore, since the distribution isn't random, there appears to be some regions in the field where finding a main sequence star is more probable.



Conclusions

After calculating cluster probabilities with both methods, we found 140 stars to be in the best four grids for main sequence and proximity. We can be confident these are cluster stars. Others either had a low main sequence score or were close to the main sequence but had a low proximity score.

To better confirm our results, we would translate our star numbers to the numbers used on WEBDA for open cluster M23. We could then compare our probabilities for each star to previously published findings.